

BUSINESS MEMO

Citi Bike × AXA Embedded Insurance Partnership

Executive Summary

This memo summarizes a data-driven partnership concept between Citi Bike (operator context: Lyft) and an insurance carrier (e.g., AXA) using (1) Citi Bike trip data to quantify exposure and rider behavior patterns, and (2) NYC crash proximity data to build a risk proxy for NYC only. The goal is to propose practical partnership launch options that improve rider trust and safety while enabling a low-friction, privacy-safe embedded insurance pilot. It also presents findings from our analysis of 121+ million Citi Bike trips (NYC) and 2.7+ million trips (Jersey City) integrated with NYPD crash proximity data.

Key Finding:

The Citi Bike system represents a large, fast-growing market with identifiable risk patterns that can inform insurance product design and targeted distribution strategies.

Important Limitation:

This analysis uses crash proximity data as a risk proxy. We do not have actual insurance claims data, so financial projections (premiums, loss ratios, profitability) would require actuarial assumptions beyond this dataset. This memo focuses on what the data does tell us.

The Problem

Before presenting the data and recommendations, it is worth being explicit about the problems this analysis is going to solve: How to add value to citibike through a cooperation for example with an insurance company.

1. What Problem Are We Solving for Each Side?

Citi Bike Riders

- **No built-in coverage:** riders who are injured in incidents have no automatic protection. This creates both a personal financial risk for the rider and a reputational and support burden for Citi Bike and Lyft when incidents escalate publicly [1].

Operators — Citi Bike / Lyft

- **Safety prioritization:** Lyft needs a structured, data-driven framework to identify where and when incidents are most likely, so that resources, staffing, messaging, interventions can be deployed efficiently rather than reactively [1].
- **Shock playbook:** COVID-like demand disruptions exposed the absence of an operational protocol for when ridership patterns break sharply [1].
- **Structural operational challenges:**

Citi Bike and Lyft face ongoing pressure from rebalancing failure (bikes unavailable where demand exists), Ghost Docks (docking stations with no available bikes or empty slots), and the absence of public subsidy that most transit systems benefit from. Because the system offers no built-in insurance or liability coverage for riders, users are left personally exposed to medical and legal costs after incidents, further deepening the friction between the operator and its community. These create cost and trust pressures that a well designed insurance partnership can partially offset by improving rider confidence and reducing post-incident support load [1, 2].

[1]: [Plos](#), [PubMedCent](#), [DavidoffLow](#)

[2]: [ibo](#), [NYC DOT](#), [Columbia.edu](#)

Insurers e.g., AXA

- **Controlled distribution:** Insurers need a high-volume, predictable distribution channel with low customer acquisition cost. Citi Bike's annual trips represent exactly that.
- **Embedded insurance:** Embedded insurance is expanding beyond consumer markets, expected to grow with over 30% of all insurance transactions likely occurring through embedded channels by 2028 [3].

Data Sources & Methodology

Data We Have (Validated Findings)

- **Trip exposure data (NYC + Jersey City):** trip volumes enabling analysis across year / month / day-of-week / hour and station-level concentration.
- **NYC crash proximity dataset (NYPD-based):** used to compute a station-month level risk proxy as crashes near station / trips, smoothed with Empirical Bayes (EB) to reduce noisy stations. This analysis has been done for two different radius 500 and 100 m, neighbourhood and dock level respectively
- **Station-month unit of analysis:** aligns with operational decision-making (seasonality, staffing, targeted interventions) while keeping the risk signal interpretable. This unit of analysis can be adjusted based on stakeholder expectations.

Data We Do NOT Have

Missing Data	Impact on Analysis
Insurance claims history	Cannot calculate actual loss ratios
Premium pricing tests	Cannot validate willingness-to-pay
Attach rate data	Cannot predict adoption rates
Citi Bike-specific crashes	NYPD data is all cyclists, not just Citi Bike

Finding 1: Market Size & Growth

Market	2019	2025	Growth
New York City	20,551,697	45,761,685	+123%
Jersey City	404,895	994,941	+146%
Combined	20,956,592	46,756,626	+123%

Insight:

The market has more than doubled since 2019, demonstrating strong secular growth beyond COVID recovery. Each trip represents a potential insurance transaction.

[3]: [BCG](#), [Zurich Insurance](#)

COVID-19 Impact & Recovery

Metric	NYC	Jersey City	Implication
2020 Decline	-4.9%	-20.2%	JC more commuter-dependent
Recovery Year	2023	2023	Both fully recovered
2025 vs 2019	+123%	+146%	JC growing faster

COVID-19 provides a critical "stress test" lens: mobility patterns can shift rapidly, and a partnership must remain operationally and reputationally safe under shocks. NYC recorded a low of 19.6M trips in 2020 (April dipping to 684K), before recovering to 35.1M in 2023 and surging to 45.8M in 2025.

Operational Takeaway for Partnership Design

- **Build a Disruption Mode** that prioritizes support-first UX (incident guidance, clarity, stability) and avoids "fear-selling" when ridership patterns break.
- **Use shocks as a governance trigger:** temporary reduction of upsell frequency, stable pricing, increased support routing.

Finding 2: Temporal Patterns

Time Window	NYC Share	JC Share
Peak Season (Jun-Oct)	54.4%	54.0%
Peak Month (September)	11.6%	10.9%
Weekdays (Mon-Fri)	73%	72%
Tue-Wed Peak	30.1%	29.7%

Insight:

Both markets show nearly identical temporal patterns. This means a single marketing calendar and product structure could serve both markets efficiently.

Idea pitch to NYC the same can apply to JC

Daily (Hour-of-Day) Breakdown

- **Morning rush (7–9am, weekdays):** sharp commuter peak- highest intent for safety nudges and concierge visibility.
- **Midday (10am–3pm):** moderate, steady; weekends trend higher- best window for optional protection attach rate (casual riders).
- **Evening peak (4–7pm, highest volumes at ~5–6pm):** highest daily exposure. The 5pm hour accounts for 7.3% of daily volume (11.3M annual trips). Strongest safety nudges and Incident Concierge entry point.
- **Night (8pm–6am):** low volume; higher safety sensitivity. Support-first messaging only; minimize upsell prompts.

Seasonal (Month-of-Year) Breakdown

- **Winter trough (Jan–Feb):** 1–2M trips/month (weather-dependent). Use for product QA and low-distraction experiments.
- **Spring ramp (Mar–May):** 3–4M trips/month. Ramp placement and messaging tests.
- **Summer peak (Jun–Sep):** 4.5–5.3M trips/month (~54% of annual volume); optimal riding conditions. Scale the pilot here for fastest learning. September 2025 set a record at 5.3M trips.
- **Fall decline (Oct–Dec):** 2–4.7M trips/month. Consolidate + seasonal safety messaging (darkness / weather awareness).

Day-of-Week Patterns

- Weekdays dominate at 71–73% of all trips across years. Tuesday and Wednesday consistently lead, each capturing ~15% of weekly volume.
- Weekend ridership shows distinct leisure patterns with more evenly distributed hourly usage better for optional protection offer placement.

Business meaning: we can schedule safety nudges, concierge staffing, and experiments by "known" high-exposure windows, reducing both cost and noise in measurement.

Finding 3: Risk Analysis (NYC Only)

We used Poisson CI for raw risk rate and Empirical Bayes (EB) smoothing to estimate station-month risk based on crash proximity normalized by trip exposure. Due to rebalancing citibike problems the exposure here is touchpoint. The EB formula is: $r_{EB} = (\text{Crashes} + \text{Baseline_Rate} \times m) / (\text{Trips} + m)$, where m is the prior strength, auto-calibrated, exposure weighted and bounded [1k–100k]. To calculate m we used Method Of the Moment (MOM). This produces a stable prioritization mechanism that avoids overreacting to small denominators and prevents outliers from distorting rankings.

Risk Trends Over Time

- The risk proxy¹ improved materially since 2019 down ~67% per trip. This finding is a match with New York City Department of Transportation (NYC DOT)²
- Risk is higher in winter months and lower in summer (counter-intuitively), likely driven by seasonal exposure denominator effects and weather-driven behavioral changes.

Station Classification — Four-Zone Model

We classify all station-months (with ≥5,000 trips for credibility) into four operational zones:

¹This should be addressed carefully as a "proxy trend," not a claims-confirmed safety improvement.

²Since 2019, New York City has moved from a pre-pandemic baseline through a significant "pandemic spike" in traffic violence, eventually reaching historic lows by 2025 and early 2026 [NYC.gov](https://www.nyc.gov)

Zone 1 — Safety Hotspot	Zone 2 — Targeted Prevention	Zone 3 — Growth Engine	Zone 4 — Control
High Exposure / High Risk Strongest safety nudges + Incident Concierge. Primary pilot target.	Lower Exposure / High Risk Targeted Prevention: focused safety messaging and early-warning monitoring for hotspot escalation.	High Exposure / Lower Risk Growth Engine: best zone for opt-in protection upsell, acquisition campaigns, and A/B testing.	Lower Exposure / Lower Risk Control & Optimization: stable baseline for measurement, holdout, and low-priority investment.

Zone 1 Anchor Stations — Persistent Hotspots

The following five stations have appeared in Zone 1 (High Exposure + High Risk) consistently across 2020, 2023, and 2025. Their persistence across years confirms structural hazard, not statistical noise. These are the recommended starting points for the pilot to avoid one-off noise.

Station Name	Total Trips	Avg. Yearly Risk Rate (per 100k trips)	Years in Zone 1
Forsyth St & Broome St	178,841	565.83	2020, 2023, 2025
Howard St & Lafayette St	172,806	478.38	2020, 2023, 2025
W 42 St & 8 Ave	284,883	477.25	2020, 2023, 2025
E 45 St & 3 Ave	74,076	465.51	2020, 2023, 2025
E 65 St & 2 Ave	103,430	429.91	2020, 2023, 2025

Note: both a 500m radius and a 100m radius crash proximity analysis were conducted. Forsyth St & Broome St and W 42 St & 8 Ave appear as high-priority targets at both radii, confirming their robustness as intervention sites.

Important Caveat:

NYPD crash data includes all cyclists, not just Citi Bike riders. This is a proxy for risk, not a direct measurement of Citi Bike incidents.

Finding 4: Jersey City — Exposure-Only Analysis

JC is best positioned as a distribution and learning accelerator, not a risk-scored geography until NJ crash/injury sources are integrated. The strongest actionable insight is station concentration.

- **Top 5 stations ≈ 67% of JC trips.** A small operational footprint can power a high-signal pilot with minimal complexity.
- **Grove St PATH dominates:** 192,000 annual trips (2025), appearing as the top station consistently across all years analyzed.
- **Strong weekday commuter pattern:** Friday leads at 15.7% of weekly trips; weekdays account for 74% of total JC volume.

- **Hourly pattern:** 5–6pm evening peak with 267K and 266K trips respectively; 8am morning rush at 199K nearly identical to NYC commuter behavior.
- **Business meaning:** a small number of stations and time windows can power a high-signal pilot with minimal operational complexity.

Recommendations

1. Partnership Options

Five plans were evaluated. All five are viable depending on stakeholder priorities:

#	Plan Name	Description
1 ★	4-Zone Safety + Support + Optional Protection	Flagship balanced plan. Safety nudges in hotspots, incident concierge as trust layer, optional micro-protection at checkout. Zones define where to prioritize safety vs. scaling. RECOMMENDED.
2	Zone 1 First + Safety Fund Loop to have city support	Serious safety + city politics plan. Lead with Zone 1 hotspot remediation; position insurance as funding safety improvements and rider recovery support.
3	Concierge-First Wedge, then Insurance	Internal champion plan. Launch concierge first (ops value + trust), then add opt-in protection once the flow is proven.
4	Zone 3 Growth Engine + Zone 1 Safety Guardrail	Revenue-first but safe. Drive conversion primarily in high-exposure/lower-risk areas; keep Zone 1 as visible safety legitimacy layer.
5	Corporate / Campus Bundle + 4-Zone Safety Reporting	Stable underwriting plan. B2B bundle with group coverage and quarterly zone-based safety reporting.

2. Primary Recommendation: Plan 1

Plan 1 (balanced) would be the best "winning partnership" for a pilot because it:

- Uses all available signals (NYC risk proxy + NYC/JC exposure patterns) without over-claiming certainty.
- Leads with safety + support in Zone 1, protecting brand reputation and city government relationships.
- Scales opt-in protection where it is safest and most measurable, Zone 2 (Growth Engine) and JC concentration.
- Is compatible with responsible data use (minimal sharing, privacy-by-design).

Pilot Plan — Operating Calendar & Disruption Mode

Pilot Scope (8–12 Weeks)

Geographies: NYC (risk + exposure scoring) + JC (exposure-only scale).

Station Selection

- **NYC:** start with Zone 1 anchors (Forsyth St & Broome St, Howard St & Lafayette St, W 42 St & 8 Ave, E 45 St & 3 Ave, E 65 St & 2 Ave) + matched Zone 2 high-exposure stations.

- **JC:** top exposure hubs primarily Grove St PATH and the next 4 stations that collectively represent ~67% of JC volume.

What Ships — Minimum Viable Build

- **Safety Nudges:** in-app, targeted by zone and time window. Zone 1 during peak hours = most prominent. Zone 2 = lighter touch.
- **Incident Concierge:** "Had a crash? Steps + resources + claim link." This is the post-ride / post-incident trust-building layer.
- **Optional Protection Add-On:** at Single Ride / Day Pass checkout. Framed as support, not fear. Shown most prominently in Zone 2 for optimal conversion.

Operating Calendar

- **Jan–Feb:** product QA + experiments (low volume, low distraction).
- **Mar–May:** ramp placement and messaging tests.
- **Jun–Sep:** scale — peak exposure; fastest learning. This is the core pilot window.
- **Oct–Dec:** consolidate + seasonal safety messaging (darkness / weather).

Daily Scheduling

- **Weekday 7–9am & 4–7pm:** strongest safety + concierge visibility in Zone 1.
- **Midday (weekends):** best window for optional protection attach rate (casual riders).
- **Night (8pm–6am):** support-first only; minimize upsell prompts.

Disruption Mode Rules — COVID / Global Disaster Playbook

Trigger when demand deviates sharply from seasonal baseline for a sustained period, or during declared emergencies:

- **Concierge up:** make the support flow more visible and faster.
- **Sales down:** reduce upsell frequency; keep pricing stable and simple.
- **Support-first messaging:** safety guidance and service updates replace promotional content.
- **Measurement shift:** compare against "shock baseline," not pre-shock seasonality.

KPIs, Kill Switches & Minimal Data Request

Pilot KPIs (Full Framework)

Category	KPI	Goal / Note
Commercial	Attach rate (by station, zone, time window)	Track by segment; compare pilot vs. control
Commercial	Incremental revenue per ride / pass	Pilot vs. control comparison
Experience / Ops	Incident-related support tickets per 10k rides	Goal: reduce over time
Experience / Ops	Complaint rate related to offer messaging	Goal: low / flat
Insurance Learning	Early claim frequency per 10k rides	Directional

Category	KPI	Goal / Note
Insurance Learning	Time-to-pay and claim satisfaction	Process quality signal
Risk Proxy (NYC)	EB risk proxy per 100k trips by station-month	Track zone migrations over time
Risk Proxy (NYC)	Zone migration (stations moving into/out of Zone 1)	Early hotspot escalation signal
Risk Proxy (NYC)	Safety hotspot burden concentration (top zone share)	Monitor reputational risk concentration

Kill Switches — Set Jointly Before Pilot Launch

- **Checkout conversion drop >1–2% sustained** in the pilot cohort → pause and investigate.
- **Complaint rate spike** above the agreed threshold related to offer messaging → pause and investigate.
- **Any reputational red flag** (e.g., "fear-selling" perception in press, social, or city government) → immediate pause and joint review.

Minimal Data Request — Privacy-by-Design

Data Needed for Pilot

- **Aggregated exposure:** trips / minutes by station-week or station-month (+ bike type if available).
- **Proof-of-ride verification token endpoint:** trip_id + timestamp + bike type → valid / invalid.
- **Aggregated incident / support counts:** optional but valuable for calibration.

Data NOT Needed for Pilot

- No rider-level continuous location trails.
- No full individual trip histories shared with the insurer.
- No dynamic pricing inputs, price stays uniform; only messaging and placement vary.

Governance

- **Weekly joint steering group:** product + ops + legal/privacy + insurance claims lead.
- **Clear responsibility split:** Citi Bike / Lyft owns UX surfaces and messaging; insurance owns underwriting and claims operations.

Conclusion

A Citi Bike × insurer partnership can be launched credibly without claims data by using (1) exposure behavior patterns to define when and where to act, (2) an NYC risk proxy to prioritize safety hotspots responsibly, and (3) privacy-safe verification to keep data sharing minimal.

The recommended Plan 1 pilot leads with safety and rider support in Zone 1, scales optional protection in Zone 3 and JC hubs, and includes a Disruption Mode playbook to remain trustworthy during shocks. The five Zone 1 anchor stations persistent hotspots across 2020, 2023, and 2025 provide a structurally validated, low-noise starting point. JC's extreme station concentration (67% of volume in 5 stations) enables fast, controlled learning in parallel.

Key Takeaways

- **Large, growing addressable market:** ~46.8M annual trips with ~8–10% annual growth (NYC: 45.8M trips in 2025, a 134% recovery from the pandemic low; JC: 995K trips, up 146% since 2019).
- **Predictable activation windows:** demand concentrates in Jun–Sep peak and commuter-heavy weekday peaks; patterns are stable enough to operationalize.
- **Actionable safety targeting:** a four-zone (Risk × Exposure) framework separates Safety Hotspots from Growth Engines, reducing reputational risk and improving measurement.
- **JC is a scale lever (exposure-only):** extremely concentrated demand (top 5 stations ~67% of JC trips), making it ideal for controlled distribution and fast learning even without crash data.
- **Recommendation:** launch a balanced "Safety + Support + Optional Protection" pilot that (a) leads with safety and incident support in the highest-impact NYC hotspots and (b) uses high-exposure, lower-risk areas and JC concentration for scalable opt-in protection testing. A "Disruption Mode" playbook (informed by COVID-era demand shifts) ensures the program remains support-first during shocks.